Units of mass

1 Choose the most appropriate unit of measure for the following:

- Grams
- Kilograms
- Tonnes
- Milligrams

a Cup of flour

b Grain of sand

c Box of 50 oranges

d Biscuit

e Car

f Pencil case

g Your weight

h Aeroplane

2 Convert the following to milligrams (mg):

a $6\ \text{g}$ **b** $0.006\ \text{g}$ **c** $0.47\ \text{g}$ **d** $72.668\ \text{g}$

3 Convert the following to grams (g):

a $87\ \text{mg}$ **b** $155\ \text{mg}$ **c** $8505\ \text{mg}$ **d** $52\,005\ \text{mg}$

4 Convert the following to the indicated units:

a $6.251\ \text{kg}$ to grams

b $12\ \text{kg}$ into grams

c $946\ \text{g}$ to kilograms

d $11\ \text{t}$ into kilograms

e $3000\ \text{g}$ into kilograms

f $5\ \text{t}$ into kilograms

g $9.272\ \text{t}$ to kilograms

h $0.493\ \text{t}$ to kilograms

i $1000\ \text{kg}$ into tonnes

j $8500\ \text{kg}$ to tonnes

5 Convert the following to the indicated units:

a $7.538\ \text{kg}$ to milligrams

b $0.75\ \text{t}$ into grams

c $0.054\ \text{kg}$ to milligrams

d $11\,540\,000\ \text{g}$ into tonnes

Estimation

6 State a reasonable estimate for the mass of each of the following:

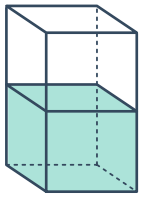
a Goat **b** Cat **c** 50 dollar bill **d** Can of soup

e Standard die **f** Newborn baby **g** Cherry **h** Black rhinoceros

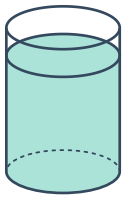
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

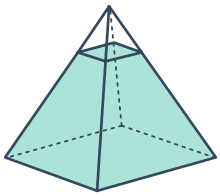
7 The container shown has 4 mg of jelly in it. Estimate the weight of the container when it is full.



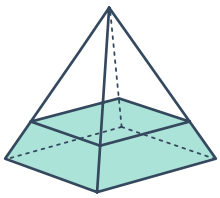
8 The container shown has 12 mg of water in it. Estimate the weight of the container when it is full.



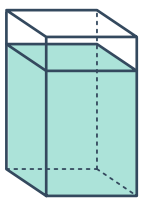
9 The container shown has 9 kg of liquid in it. Estimate the weight of the container when it is full.



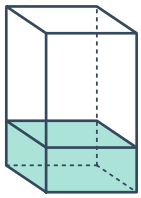
10 The container shown has 12 g of water in it. Estimate the weight of the container when it is full.



11 The container shown has 15 g of liquid in it. Estimate the weight of the container if it was full.



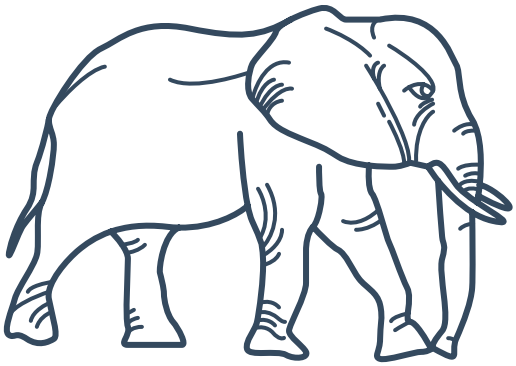
12 The container shown has 1 g of jelly in it. Estimate the weight of the container when it is full.



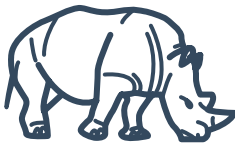
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 An elephant has a mass of $5000\ \text{kg}$:

Elephant



Rhinoceros



Use the diagram to estimate the mass of a rhinoceros.

14 A tablet has a mass of 640 grams:





Use the diagram to estimate the mass of a laptop.

Applications

15 Adam's toy truck has a mass of $200\ \text{g}$. Find the weight of the toy truck in milligrams.

16 Valerie has fruit everyday for dinner. Today she consumed 45 cherries, which weighed a total of $180\ \text{g}$. How many milligrams did the average cherry weigh?

17 Rewrite $2468\ \text{g}$ in terms of the number of whole kilograms (kg) and grams (g) left over.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 Jordan's bag weighs $\frac{1}{2}$ of a kilogram. How many grams does the bag weigh?

19 Find the total mass of 12 cans of soup, in kilograms, if each has a mass of $350\ \text{g}$.

20 The cost of mining gold for a mining company is $\$840$ per gram. How much will it cost the company to mine $0.2\ \text{kg}$ of gold?

21 Justin is weighing himself and his dog at the same time. Their combined mass is $70\ \text{kg}$. Justin knows that he weighs $32\ \text{kg}$, so what must his dog weigh?

22 Mohamad's trampoline can hold a maximum of $90\ \text{kg}$ of mass. If Mohamad has some friends over and they each weigh $30\ \text{kg}$, how many of them can use the trampoline at once?

23 At a sporting match, the coach has 10 oranges in a container to give out to the team members at half-time. The container of oranges weighs $800\ \text{g}$. How much does each orange weigh?

24 If 11 apples have a total mass of $3.19\ \text{kg}$, find the average mass of those apples in grams.

25 A truck is transporting 8 horses. Each horse in the truck weighs $800\ \text{kg}$.

- **a** How much do all the horses in the truck weigh altogether?
- **b** If the truck weighed $4000\ \text{kg}$ before the horses were loaded into it, how many kilograms does the loaded truck weigh now?

26 James buys a $4000\ \text{g}$ box of washing powder. If he uses $2.3\ \text{kg}$ of the washing powder, how many grams of powder is left in the box?

27 Christa buys a $8\ \text{kg}$ bag of rice. If she uses $2800\ \text{g}$, how many kilograms of rice is left?

28 Passengers have a limit of $20\ \text{kg}$ of luggage on a plane. If James's bag was over the weight limit by $1800\ \text{g}$, how much does his bag weigh in kilograms?

29 If each carrot has a mass of $70\ \text{g}$ and each lemon has a mass of $60\ \text{g}$, how many kilograms would 6 carrots and 9 lemons weigh altogether?

30 If a truck can carry $720\ \text{kg}$ of soil in one load, how many loads does it need to take in order to empty a field of $23.04\ \text{t}$ of soil?

31 The nutritional information on a brand of corn chips states that there is $6\ \text{mg}$ of sodium in every $100\ \text{g}$ of corn chips. How many grams of sodium would there be in 10 packets of corn

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

32 If 25 bags of flour have a mass of 2 kg, find the mass of 5 bags, in grams.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647